

Software Presentation: Jenkins

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What is Jenkins?

- **Open-source automation server.**
- **Helps automate building, testing, and deploying code.**
- **Supports continuous integration (CI) and continuous delivery (CD) processes.**
- **Works with many programming languages, including Python.**
- **Helps detect issues early and speeds up development cycles.**

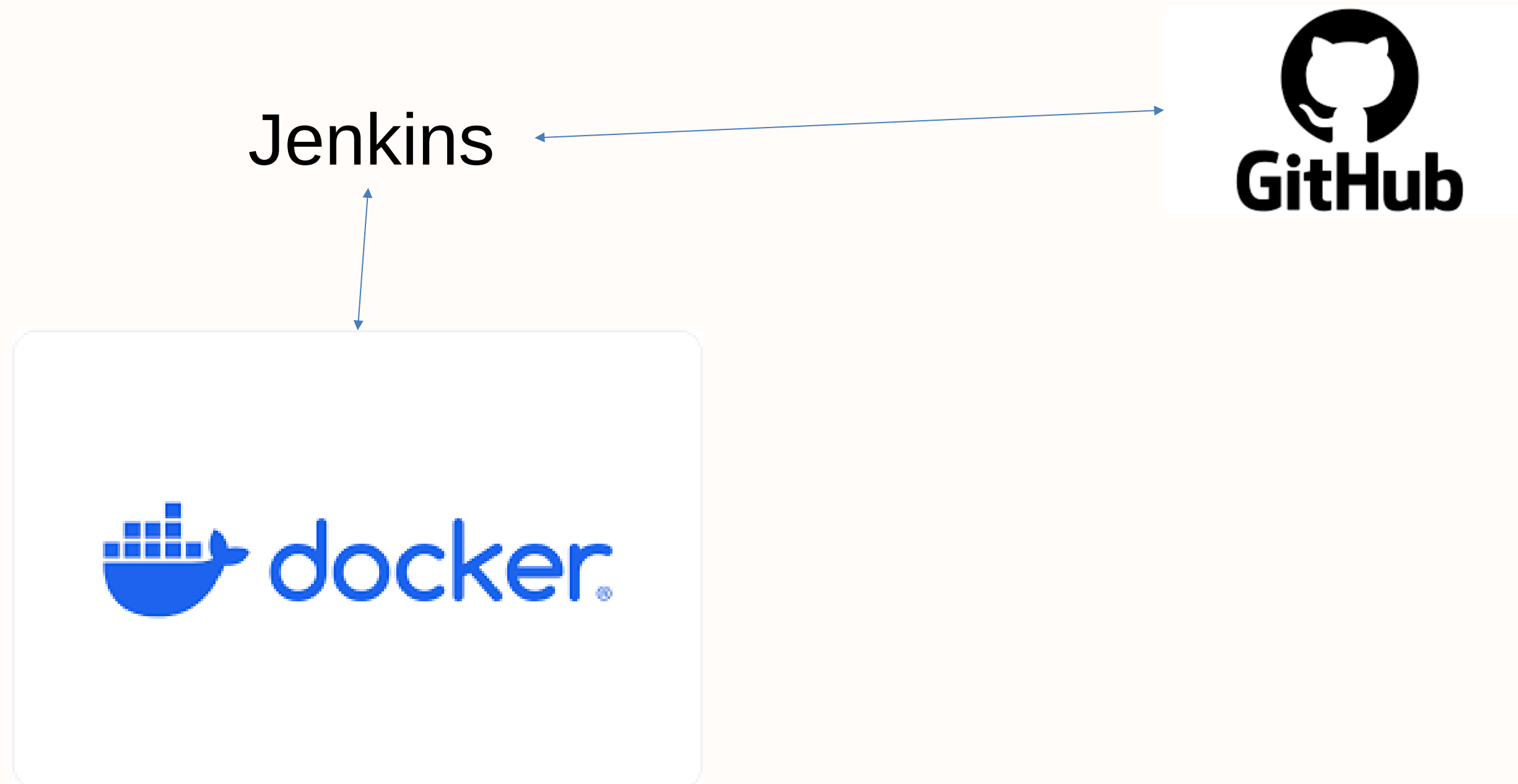
What is CI/CD?

- **Stands for Continuous Integration and Continuous Deployment**
- **Continuous Integration (CI):** Automatically tests and integrates new code changes into a shared repository.
- **Continuous Deployment (CD):** Automatically deploys code to production without manual steps as well as passing test cases.
- **Improves code quality, reduces bugs, and speeds up release cycles.**
- **Essential for modern DevOps and agile development practices.**

Prerequisites

- **Docker**
- **Linux commands**
- **Python**
- **Json/Yaml**
- **Shell/Bash commands**

Jenkins Architecture



Jenkins Installation

<https://www.jenkins.io/doc/book/installing/>

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Installing Jenkins

The procedures in this chapter are for new installations of Jenkins.

Jenkins is typically run as a standalone application in its own process. The Jenkins WAR file bundles [Winstone](#), a [Jetty](#) servlet container wrapper, and can be started on any operating system or platform with a version of Java supported by Jenkins.

Theoretically, Jenkins can also be run as a servlet in a traditional servlet container like [Apache Tomcat](#) or [WildFly](#), but in practice this is largely untested and there are many caveats. In particular, support for WebSocket agents is only implemented for the Jetty servlet container. See the [Servlet Container Support Policy](#) page for details.

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Docker ⇒

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Build the Jenkins Image via Docker

Git clone https://github.com/yaseen-ghani/DSC_672_jenkins_demo

```
docker build -t myjenkins-blueocean:2.492.3-1 .
```

```
docker network create Jenkins
```

```
docker run --name jenkins-blueocean --restart=on-failure --detach `
--network Jenkins --env DOCKER_HOST=tcp://docker:2376 `
--env DOCKER_CERT_PATH=/certs/client --env DOCKER_TLS_VERIFY=1 `
--volume jenkins-data:/var/jenkins_home ` jenkins-blueocean
--volume jenkins-docker-certs:/certs/client:ro `
--publish 8080:8080 --publish 50000:50000 myjenkins-blueocean:2.492.3-1
```

```
docker exec jenkins-blueocean cat /var/jenkins_home/secrets/initialAdminPassword
```

Jenkins url: <https://localhost:8080/>

Docker Node Creation

```
docker run -d --restart=always --network Jenkins -p 2375:2375 -v  
/var/run/docker.sock:/var/run/docker.sock alpine/socat tcp-listen:2375,fork,reuseaddr unix-  
connect:/var/run/docker.sock
```

```
docker ps
```

```
docker inspect <container_id>
```

```
jenkins/agent:alpine-jdk21
```


Jenkins Pipeline

```
pipeline {
  agent {
    node {
      label 'docker-alpine'
    }
  }
  triggers {
    pollSCM '* * * * *'
  }
  stages {
    stage('Build') {
      steps {
        echo "Building.."
        sh ""
        echo "doing building stuff.."
        ""
      }
    }
    stage('Test') {
      steps {
        echo "Testing.."
        sh ""
        echo "doing testing stuff.."
        ""
      }
    }
    stage('Deliver') {
      steps {
        echo 'Deliver....'
        sh ""
        echo "doing delivery stuff.."
        ""
      }
    }
  }
}
```